



F1-10 Hydraulics Bench

The F1-10 unit is a portable and self-contained service module providing a controlled flow of water to a range of optional accessories. It is supplied as standard with the Fluid Mechanics F1-aBASIC Software

PRODUCT CODE: F1-10

Categories: Agricultural, Civil, Educational, F1 Series Fluid Mechanics, F1-10 Hydraulics Bench

Tags: Basic Fluid Mechanics , Complete Fluid Mechanics Laboratory , Corrosion-Resistant , Electronic Flow Meter , F1-10 , F1-ABASIC Educational Software , Hydraulics Bench , Inline Digital Flow Meter , Mobile Bench , Open Channel , Quick Release Pipe Connectors , Reduce Turbulence , Small Flow Rates , Stilling Baffle , Submersible Pump , Volumetric Measuring Tank

DESCRIPTION

The F1-10 unit is a portable and self-contained service module providing a controlled flow of water to a range of optional accessories. It is supplied as standard with the **Fluid Mechanics F1-aBASIC Software**.

This mobile bench is constructed from lightweight corrosion-resistant plastic and incorporates an open channel with side channels to support the accessories on test. The hydraulics bench includes a volumetric measuring tank stepped to accommodate low or high flow rates and a stilling baffle to reduce turbulence. A remote sight tube with scale gives an instantaneous indication of water level.

The bench additionally includes a quick-release pipe connector situated in the benchtop enabling rapid exchange of accessories without the need for hand tools, a measuring cylinder for measurement of very small flow rates, stopwatch and a copy of **Armfield's F1-aBASIC educational software**.

The F1-10 hydraulics bench can be supplied with either a factory fitted electronic flow meter with digital display or an optional inline digital flow meter that can be added in line to the experiment on test at any time.

TECHNICAL SPECIFICATIONS

Circulating Pump:

- Submersible
- Max head: 8.3m H₂O
- Max flow: 80 litres/min (using volumetric tank)
- Max flow: 100 litres/min (using appropriate accessory)

Motor rating: 0.25kW
Sump tank capacity: 250 litres
High flow volumetric tank: 40 litres
Low flow volumetric tank: 6 litres
Height of working surface: 1m above floor level

FEATURES & BENEFITS

Included Software armSOFT F1-aBASIC

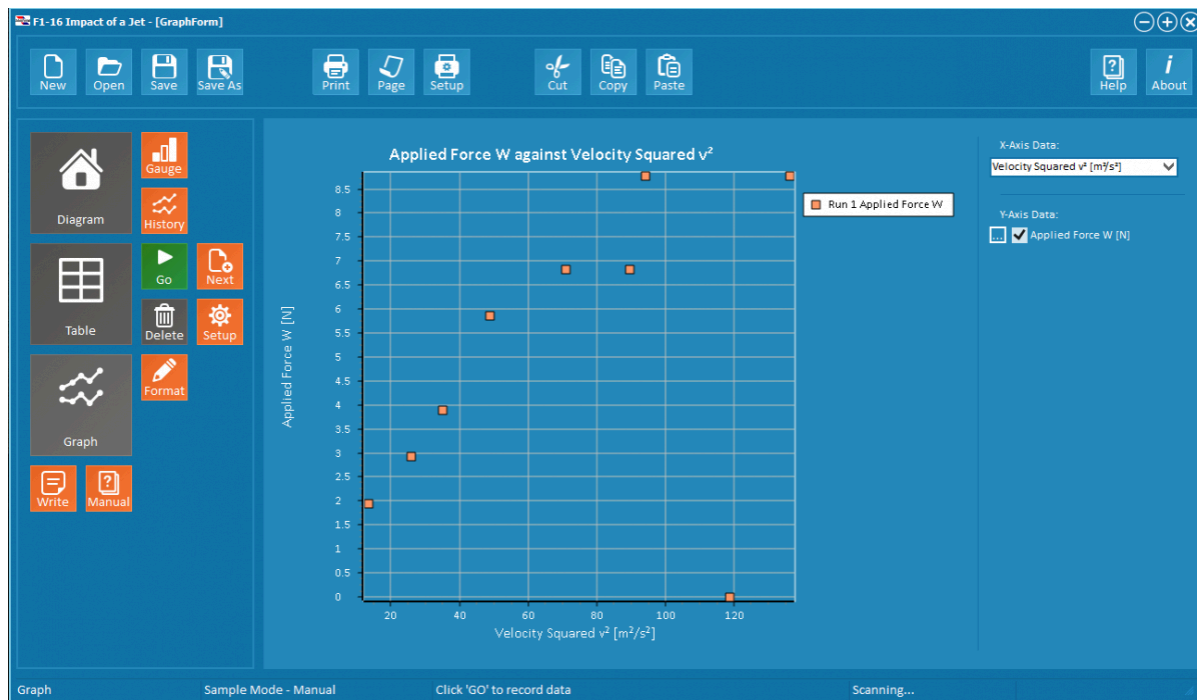
Armfield's F1-aBASIC software is now included as standard with either of the hydraulic benches. The Armfield software is a powerful manual data entry learning package which enhances the educational content and understanding of Armfield's F1 Fluid Mechanics accessories that utilise either of the F1-10 Hydraulics benches.

The software allows the user to manually input data from primary instrumentation and offers a powerful tool for displaying and processing the results.

Software additionally includes the electronic version of the manual for all the modules on test.

Some of the major features include:

- **Graphical Display** – When several samples have been recorded, they can be viewed in graphical format. Powerful and flexible graph plotting tools are available in the software allowing the user full choice over what is displayed
- **Mimic Diagram** – a pictorial representation of the equipment with fields to enter measurements from the equipment which displays any calculated variables directly in engineering units
- **Tabular Display** – As the data is entered, it is stored in a spreadsheet format. The table also contains columns for the calculated values



F1-16-MKII Impact of a Jet Graphical Display

F1-16-MKII Impact of a Jet mimic Diagram

F1-16-MKII Impact of a Jet Tabular Display

Requirements

Electrical supply:

Basic Hydraulics Bench (with manometer):

1. F1-10-A 220-240V / 1ph / 50Hz @ 10 amp
2. F1-10-B 110-120V / 1ph / 60Hz @ 20 amp
3. F1-10-G 220V / 1ph / 60Hz @ 10 amp

Digital Hydraulics Bench (with digital flow meter & manometer):

1. F1-10-2-A 220-240V / 1ph / 50Hz @ 10 amp

2. F1-10-2-B 110-120V / 1ph / 60Hz @ 20 amp
3. F1-10-2-G 220V / 1ph / 60Hz @ 10 amp

F1-10-1 Digital Flow Meter for F1-10 Hydraulics Bench (attaches to water outlet and can be removed as necessary):

1. 110-120V / 1ph / 60Hz @ 20 amp

Water: Fill with clean water. No permanent connection required.

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 1.5m³

Gross Weight: 160Kg

Dimensions

Length: 1.13m

Width: 0.73m

Height: 1.0m

Order Codes

Basic Hydraulics Bench:

- F1-10-A 220-240V / 1ph / 50Hz @ 10 amp
- F1-10-B 110-120V / 1ph / 60Hz @ 20 amp
- F1-10-G 220V / 1ph / 60Hz @ 10 amp

Digital Hydraulics Bench (with digital flow meter):

- F1-10-2-A 220-240V / 1ph / 50Hz @ 10 amp
- F1-10-2-B 110-120V / 1ph / 60Hz @ 20 amp
- F1-10-2-G 220V / 1ph / 60Hz @ 10 amp

F1-10-1 Digital Flow Meter for F1-10 Hydraulics Bench

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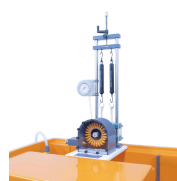
**F1-13-MKII / F1-
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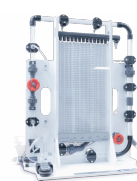
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